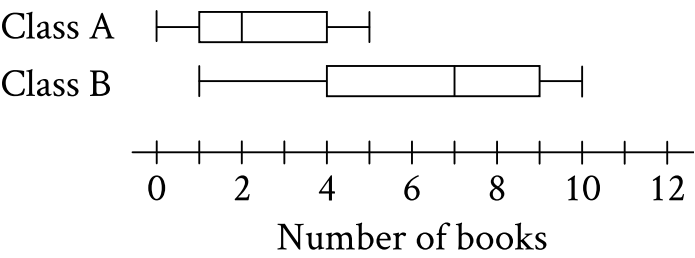


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



The two box plots show the distribution of number of books read over the summer by the students in two different English classes. What is the positive difference between the ranges of number of books read over the summer for the two classes?

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that the two boxplots show the distribution of number of books read over the summer by the students in two different English classes. In a boxplot, the first vertical line represents the minimum value of the data set and the last vertical line represents the maximum value of the data set. The range of a data set is the difference between its maximum value and its minimum value. In class A, the maximum number of books read is **5** and the minimum number of books read is **0**. The difference between those values is **5 — 0**, or **5**. Therefore, the range of the number of books read in class A is **5**. In class B, the maximum number of books read is **10** and the minimum number of books read is **1**. The difference between those values is **10 — 1**, or **9**. Therefore, the range of the number of books read in class B is **9**. To find the positive difference between the ranges of the number of books read for the two classes, the smaller range must be subtracted from the larger range. Therefore, the positive difference between the ranges of number of books read over the summer for the two classes is **9 — 5**, or **4**.